

MADHUKAR GOLI

AI ML Systems Engineer

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Open to Relocation

PROFESSIONAL SUMMARY

AI Systems Engineer with **7+ years** of experience designing enterprise-grade **RAG pipelines**, **agentic AI systems**, and LLM-powered applications. Specialized in advanced document processing, hallucination mitigation, multi-agent orchestration, and scalable AI microservices. Track record of reducing hallucination rates, improving retrieval accuracy, and deploying production AI systems serving enterprise workloads at S&P Global and in independent product development.

CORE TECHNICAL SKILLS

LLM Frameworks & Orchestration: LangChain, LangGraph, Prompt context Engineering, Conversational Memory, Guardrails

RAG & Retrieval Systems: Advanced & Agentic RAG, Hybrid & Multi-Vector Retrieval, Semantic Search, Hierarchical & TOC-Aware Chunking, Metadata-Driven Indexing, Grounding, Retrieval Validation, Validation loops, Encoder models, Reranking

Agentic AI: Crew AI, Model Context Protocol(MCP), Multi-Agent Systems, Planning & Task Decomposition, Autonomous reasoning & Agent Orchestration, Supervisor-Worker Architecture, Human-in-the-Loop (HITL)Tool / Function Calling Pipelines, Memory & Routing, Retry & Fallback Strategies

AI Reliability: Hallucination Mitigation, Confidence Scoring, LLM Evaluation & Metrics, A/B Testing, Latency Profiling

Vector Databases: Redis Search, Qdrant, ChromaDB, Pinecone, Weaviate, Milvus, FAISS, Elasticsearch, Vespa, pgvector

Backend & APIs: Python, DSA, PostgreSQL, NoSQL, FastAPI(REST), YAML/JSON Schema, Open APIs, Hugging Face, Llama

Cloud & DevOps: AWS (Amazon Bedrock, SageMaker AI, EKS), Azure (Azure AI Foundry), GCP (Vertex AI, GKE), Red Hat OpenShift, Docker, GitHub Actions CI/CD, Cloud-Native AI Systems

ML & Analytics: NLP Pipelines, Predictive Modeling, Feature Engineering, Time-Series Forecasting, BigQuery ML, Power BI

PROFESSIONAL EXPERIENCE

AI Systems Engineer

Jan 2026 – Present

S&P Global | Dallas, TX (Remote)

- Architected enterprise **RAG + Agentic AI platform** for large-scale regulatory and financial policy documents, enabling accuracy in query resolution across complex document corpora.
- Designed **hierarchical, TOC-aware, and metadata-driven chunking** strategy, improving retrieval accuracy across multi-section financial policies and regulatory filings.
- Built **multi-vector retrieval pipelines** using **Redis Search** with context-aware routing, reducing irrelevant chunk retrieval.
- **Eliminated hallucinations** via citation enforcement, multi-pass grounding, and validation loops, **increasing response trustworthiness** for compliance-critical use cases.
- Implemented **confidence scoring and guardrails**, improving **response reliability by ~30%** in evaluation benchmarks.
- Developed **evaluation framework** covering retrieval accuracy, grounding quality, and latency — enabling measurable iteration on system performance.
- Deployed scalable AI microservices on **RH OpenShift** are integrated with enterprise APIs, internal knowledge bases, and authentication layers.

INDEPENDENT AI PROJECTS

Lodestar Veritas – Enterprise Agentic Multimodal RAG Platform

Feb 2026 – Jun 2026

Python · LangGraph Orchestration · Ollama · Qdrant · FastAPI · Docker · Hybrid Retrieval |

<https://github.com/madhukargoli1992G/lodestar-veritas>

- Architected and developed an enterprise-inspired Agentic AI platform supporting 7+ document formats with citation-grounded document intelligence detection and workflows.
- Designed a 10+ stage agentic orchestration pipeline spanning ingestion, chunking, embeddings, hybrid retrieval, answer generation, verification, hallucination detection, confidence scoring, and retry mechanisms.
- Engineered a hybrid retrieval architecture combining Dense Search, BM25, Reciprocal Rank Fusion (RRF), and reranking to improve retrieval relevance and response grounding.
- Delivered a production-ready solution with FastAPI APIs, Streamlit UI, Dockerized deployment, Qdrant vector storage, and 90+ automated Pytest validations.

Vendor Intelligence – Copilot

Jan 2025 – Mar 2026

Independent Product Development

Python · FastAPI · Streamlit · LangChain · Qdrant · Ollama | github.com/madhukargoli1992G/vendor-intelligence-copilot

- Architected **multi-layer RAG system** for vendor contract and SLA analysis with semantic + structural chunking and hybrid retrieval pipelines. **Reduced hallucinations** using grounded responses, citation enforcement, and guardrails validated against a custom evaluation pipeline.
- Built **multi-agent reasoning workflows** with dynamic task orchestration and production-ready FastAPI microservices.

FurnitureFlip – Pro

Jul 2025 – Dec 2026

Independent Product Development

LangChain · LangGraph · FastAPI · GKE · GCP · Ollama | github.com/madhukargoli1992G/FurnitureFlip-Pro

- Designed **multi-agent AI system** with **autonomous reasoning**, tool-calling pipelines, and **hybrid RAG** for contextual accuracy. Implemented **agent memory, routing, and multi-step orchestration** with validation layers and guardrails ensuring reliable outputs.

Netflix Movie Performance Analytics

Jan 2025 – May 2025

SQL · BigQuery · Data Modeling · Data Visualization · Grafana · Predictive Analytics · KPI Design · Dashboard Development

- Engineered an **end-to-end analytics pipeline** using BigQuery and SQL to process 5,000+ movie records, deriving KPIs such as ROI, profitability, and genre performance for strategic insights. Designed **interactive Grafana dashboards** to visualize trends in ratings, runtime, and revenue, enabling data-driven content investment decisions and uncovering high-performing genres.

AI-Driven Movie Success Prediction & Piracy Impact Analytics

Aug 2025 – Dec 2025

Python · RAG · Machine Learning · NLP · Predictive Modeling · Data Analysis · Statistical Modeling · Microsoft Power BI (visualization)

- Organized **predictive models** integrating movie metadata and external factors to forecast financial success, **improving decision accuracy** for budget allocation and content strategy and **proved** by analyzing the impact of digital piracy on revenue using data-driven techniques, uncovering hidden ROI distortions and **providing actionable insights** for risk-adjusted investment planning

EDUCATION

Master of Science, Business Analytics

Aug 2024 – Dec 2025

University of the Pacific | Stockton, CA, USA

GPA: 3.84

- **Research & Teaching Assistant:** Designed LLM-driven analytical frameworks for regression outputs; built prompt orchestration pipelines; implemented hallucination guardrails improving output reliability by 30%. Mentored 40+ graduate students on Python, SQL modeling, and AI-assisted analytics.

PRIOR WORK EXPERIENCE

Senior Lead Data Scientist

Jan 2021 – Jul 2024

Kankatala Corporate | India

- Led predictive analytics initiatives forecasting demand across **100+ SKUs**, reducing stockout occurrences by **18%**.
- Enhanced time-series models with lag features and smoothing techniques, improving forecast reliability by **20%**.
- **Automated analytics workflows** using Python, cutting reporting turnaround from 2–3 days to under **6 hours**.
- Architected SQL-driven data pipelines standardizing enterprise-wide reporting and executive dashboards with promotional impact analysis.

Data Scientist

Jan 2018 – Dec 2020

Kankatala Corporate | India

- Engineered Python automation scripts reducing manual analysis effort by **32%** (6–8 hrs/week).
- Developed time-series forecasting models improving demand planning accuracy by **14–20%**.
- Optimized complex SQL queries improving reporting performance by **22%**; designed Power BI dashboards tracking 100+ SKUs across retail channels.